HH-M-351E
July 8, 1976
SUPERSEDING
Fed. Spec. HH-M-351D
November 18, 1968

FEDERAL SPECIFICATION

MILLBOARD, ASBESTOS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers one type of asbestos millboard, in flat panels, either smooth or textured surface, in a variety of thicknesses.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from the Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established points in their agencies.)

FSC 5640

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Military Specification:

MIL-W-3448 - Wallboard: Packing of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

C150 - Portland Cement.
C177 - Thermal Conductivity of Materials by Means of the Guarded Hot Plate.
E84 - Surface Burning Characteristics of Building Materials.
F36 - Compressibility and Recovery of Gasket Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Material. The millboard shall be composed of asbestos fiber and a binder composed of portland cement in accordance with ASTM C150. Vegetable matter shall be used only for sizing and shall not exceed 3 percent by weight, of the composition of the millboard. The asbestos content shall be a minimum of 60 percent (see 4.4.5).

3.2 Description. Asbestos millboard shall be flat, solid, rigid or semi-rigid sheets.

3.3 Standard commercial product. The millboard shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commerical product with any added features needed to comply with the requirements. Additional or better features which are not specifically prohibited by this specification, but which are a part of the manufacturer's standard commercial product shall be included in the millboard being furnished. A standard commercial product is a product which has been or will be sold on the commercial market through advertisements of manufacturer's catalogs or brochures, and represents the latest production model(s).

3.4 Dimensional tolerances.

3.4.1 Thickness. Unless otherwise specified (see 6.2), the thickness of the millboard shall be as shown in Table I. When the thickness of the millboard specified exceeds 1/2 inch, the same percentage of variation shall be permitted as that specified for 1/2-inch nominal thickness in Table I.

TABLE I. Thickness.

Nominal thickness	Limits	
	Minimum	Maximum
Inch	Inch	Inch
0.0625 (1/16)	0.05	0.07
0.1250 (1/8)	0.11	0.14
0.1875 (3/16)	0.16	0.20
0.2500 (1/4)	0.22	0.28
0.3750 (3/8)	0.33	0.41
0.5000 (1/2)	0.45	0.55

3.4.2 Width and length. The milboard shall be 42 inches wide and 48 or 96 inches long as specified (see 6.2). Maximum variation permitted in the overall width or length of the board shall be ±1/8 inch.

3.5 Fire rating. The millboard shall have a maximum flame-spread rating of 20. The maximum smoke-developed rating shall be 10 (see 4.4.6).

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3.6 Thermal conductivity. The millboard shall not have a thermal conductivity (k) greater than 1.50 British thermal unit in/hours square feet degrees Fahrenheit (F) at an average temperature of 212°F (see 4.4.7).

3.7 Weight. The weight of the dried millboard shall not exceed the weight shown in Table II (see 4.4.8).

TABLE II. Weights.

Nominal thickness (inch)	Maximum weight (lb/sq ft)
0.0625 (1/16)	0.375
0.1250 (1/8)	0.750
0.1875 (3/16)	1.125
0.2500 (1/4)	1.500
0.3750 (3/8)	2.25
0.5000 (1/2)	3.00

3.7.1 Loss of weight. When exposed to a temperature of 1650° to 1750°F, the loss of weight of the millboard shall not exceed 20 percent (see 4.4.9).

3.8 Tensile strength.

3.8.1 Normal. The minimum tensile strength of the millboard shall be 200 pounds per square inch (psi) (see 4.4.10).

3.8.2 After heating. The minimum tensile strength of the millboard shall be at least 170 psi after exposure to a temperature of 400°F for not less than 1 hour (see 4.4.10.1).

3.9 Resistance to change. The millboard shall show no change of texture or softening after exposure to a dry heat of 400°F for 30 minutes in a closed chamber. A hot plate is not acceptable (see 4.4.11).

3.10 Workmanship. The millboard shall be unwarped with clean, sharp unbroken edges and corners. The surfaces shall be free of dirt, checks, spalls, cracks or any defects which would impair the serviceability of the panel.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Sampling. The sample size for levels of inspection and the Acceptable Quality Level (AQL) shall be in accordance with MIL-STD-105.

4.2.1 Lot. Unless otherwise specified, a lot shall consist of all millboard of the same size submitted for inspection at the same time and produced by the same process and from the same raw material lot (see 6.2).

4.2.2 Sampling for dimensional and visual examination of end item. Sampling of millboard for dimensional and visual examination shall be in accordance with inspection level S-4.

4.2.3 Sampling for test. Sampling for test shall be in accordance with level S-2.

4.3 Visual and dimensional examination of end item. The millboard selected in accordance with 4.2.2 shall be examined for the defects listed in Table III. The AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

TABLE III. Classification of defects.

Examine	Defects	Major	Minor
End item	Length, width, and thickness not as specified.	X	
	Length, width, or thickness not within tolerance.	X	
	Cracks.	X	
	Broken edges or corners.	X	
	Surface departs from planeness.		X
	Damage or defects affecting function or serviceability.	X	
	Damage or defects not affecting function or serviceability.		X

4.4 Tests.

4.4.1 Acceptable Quality Level. The AQL for tests shall be 6.5 percent defective.

4.4.2 Length, width, and thickness. The length, width, and thickness of the boards shall be measured with a steel rule graduated in 1/16-inch increments.

4.4.3 Edges. The squareness of the edges shall be determined with the use of a carpenter's square. One leg of the carpenter's square shall be placed alongside the length of the board with the right angle of the square alined against the edge. The distance from the edge of the square to the edge of the board shall be not more than 1/8 inch per foot.

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4.4.4 Preparation of specimens. The test specimens shall be cut from the interior area of the samples in such a manner that no edge of a specimen shall be less than 3 inches from the original edges of the sample. For the maximum weight test cut a specimen 12 inches square. The specimens for the tensile test should be cut so that the length of the specimens are parallel to the lay of the fiber. The dimensions should be 1-inch wide and 12 inches long. A specimen, 6 inches square, shall be cut for the resistance-to-change test. Two 2-gram specimens shall be taken, one for loss-of-weight, and the second for the asbestos content.

4.4.4.1 Conditioning of specimens. Grind the two gram specimens and dry at 220°F to a constant weight. Then store the samples at room temperature in a desiccator containing freshly-activated anhydrous calcium sulfate desiccant.

4.4.5 Asbestos content. A certificate of compliance shall be submitted attesting the asbestos content of the millboard. When certificates of compliance are submitted the Government reserves the right to check test items to determine the validity of the certification.

4.4.6 Fire rating. Flame-spread and smoke-developed ratings shall be determined in accordance with ASTM E84 (see 3.5).

4.4.7 Thermal conductivity. Thermal conductivity or "k" factor shall be determined in accordance with ASTM C177 (see 3.6).

4.4.8 Maximum-weight test. Place the 12-inch square samples in an oven at 200°F ±5°F for 2 hours. Weigh the specimen to determine the weight. Any weight in excess of that shown in Table II shall constitute failure (see 3.7).

4.4.9 Loss-of-weight test. Condition a 2-gram sample by heating at 220°F for 30 minutes. Cool to room temperature, weigh the sample and record the weight. Place the specimen in a crucible and heat to a temperature of 1650°F to 1750°F in a muffle furnace for a period of 30 minutes. Weigh the specimen after removing from the heat and determine the weight loss by noting the difference between the original weight and the weight of the specimen after heating. The weight loss determined here will include the vegetable matter used as sizing (see 3.7.1).

4.4.10 Normal tensile test. Firmly clamp a tensile specimen squarely in the jaws of a tensile machine, with a clearance between the jaws of from 8 to 10 inches. The machine shall operate at a speed of 12 inches per minute. Determine the tensile strength at rupture in terms of psi (see 3.8.1).

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4.4.10.1 Tensile test (400°F). Place the second tensile specimen in a dry heat oven set at 400°F for 1 hour. Then store the specimen in a desiccator containing freshly-activated anhydrous calcium sulfate desiccant until cooled to room temperature. Then test the specimen as specified in 4.4.10 (see 3.8.2).

4.4.11 Resistance-to-change test. Measure the compressibility and recovery properties of a specimen in accordance with ASTM F36, Procedure H. Condition the specimen by heating in a dry atomsphere at 400°F for 30 minutes. Cool the specimen to room temperature and remeasure compressibility and recovery. Any change greater than 10 percent shall constitute failure of this test (see 3.9).

4.5 Preparation for delivery inspection. The packaging, packing, and marking shall be examined and tested to determine compliance with Section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. The millboard shall be packaged in accordance with the level A requirements of MIL-W-3448.

5.1.2 Level C. The millboard shall be packaged in accordance with the contractor's standard practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The millboard shall be packed in accordance with the level A requirements of MIL-W-3448.

5.2.2 Level B. The millboard shall be packed in accordance with the level B requirements of MIL-W-3448.

5.2.3 Level C. The millboard shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules, or with National Motor Freight Classification rules.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract, packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

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5.3.2 Military agencies. In addition to any special marking required be the contract, packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Asbestos millboard is intended for interior use to serve as temporary protection against fire (surface flame spread), heated gas, liquid or solid objects and against acids and corrosive liquids.

6.2 Ordering data. Purchasers should select the preferred options offered herein and include the following data in procurement documents:

(a) Title, number, and date of this specification.
(b) Thickness required (see 3.4.1).
(c) Length of board required (see 3.4.2).
(d) Size of lot, if different (see 4.2.1).
(e) Level of packaging and level of packing required (see 5.1 and 5.2).

6.3 Asbestos millboard should be purchased on a basis of square feet.

6.4 Dry heat. Dry heat is defined as the heat obtained in a closed chamber, such as an oven, rather than the heat obtained from a heated surface.

6.5 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Armed Services Procurement Regulation (ASPR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

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MILITARY INTEREST:

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Navy - YD
Air Force - 84

User activities:

Army - CE
Navy - MC

Civil Agency Coordinating Activities:

GSA-FSS
COMMERCE-NBS

Preparing activity:

Navy - YD

Project No. 5640-0286

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